

A code you can undo

Name: _____

Date: _____

Use cards A, B, C, D. A key uses every letter exactly once in its output row. One turn replaces **every** letter at once. Keep spaces and letter positions fixed.

Input	A	B	C	D
Key P	B	C	A	D
Key Q	B	A	D	C

1. What stays the same?

Encode ABBA with P. Could any valid key turn ABBA into ABCD? Explain. Then decode P's output CAAC.

My messages and a reason:

2. Keep coding the code.

Start with ABCD. Apply P again and again until **all four letters** are back in their starting places. Do the same with Q. Count the first positive return.

Turn	Key P	Key Q
0	ABCD	ABCD
1		
2		
3		
4		

Look closer: Follow just A through P. What happens to D? Is one letter returning enough to say the *whole machine* has reset?

Try it with objects. Explain by pointing, talking, or drawing.

Build a slow machine

3. Make a key that takes four turns.

Write each of A, B, C, D exactly once under the line. Test it with the whole message ABCD. Draw arrows to show the route of each letter.

Input	A	B	C	D
My key				

Arrow loops and my test:

4. Can four letters take six turns?

Try it. Then sort your keys by their loops. A letter that stays put is a loop of size 1. What are all the ways to split four letters into loops?

Loop sizes	First turn when all are home

Go further: Add a fifth letter. Can you build a machine that really does take six turns? Try two separate loops.

Try it with objects. Explain by pointing, talking, or drawing.